



Smart Contract Security Review

Morpho - Midnight : fixed-rate, fixed-maturity, zero-coupon lending

0 CRITICAL - 0 HIGH

Executive Summary

Severity	Total	Open	Remediation
Critical	0	0	provided
High	0	0	provided
Medium	1	1	provided
Low	7	7	provided
Informational	19	19	provided



28 active - 29 submitted - 1 withdrawn (integrity)

CATEGORY	AUDITED FILES	REVIEWER
Isolated lending (fixed-rate ZCB)	26 (src/)	Kairos Lab Security Research
ENGAGEMENT	COMMIT	STANDARD
2026-05-21 - 2026-06-05	7538c438 (contest HEAD)	OMEGA V1 - Cantina competition

Scope & Methodology

Repository

Target: morpho-org/midnight - commit 7538c438 (contest HEAD).
Scope: src/ - Midnight core, libraries, ratifiers, periphery, interfaces (26 files).

Methodology - OMEGA V1

- Capacity-chain modelling across the market & offer trust diagram.
- Line-by-line read of src/ AND all 33 Certora/CVL proofs (intended-behaviour oracle).
- F1 invariant driver: stateful fuzz of conservation/solvency invariants - 4,000,000 calls, 0 counterexamples.
- Independent adversarial validation + a doctrine-blind divergent (heretic) review.
- Every finding empirically verified (executed PoC or proof) and citation-checked char-for-char.

Deduplication (prior reviews)

Findings deduplicated against Spearbit (41c8405a) and the prior TrustSec / Sherlock baseline (6783b978).
Net-new value on the HEAD commit incl. the #944 maxRepaid change absent from prior audits.

Qualitative Analysis



Findings Summary

0

CRITICAL

0

HIGH

1

MEDIUM

7

LOW

19

INFO

Resolution status

Resolutions proposed: 28 / 28 (100%) - each finding ships a four-tier remediation:

immediate fix - structural fix - defense-in-depth - regression / variant tests.

Open: 28 Fixed: 0 Acknowledged: 0 (pre-disclosure; awaiting maintainer triage).

Headline

0 Critical / 0 High - corroborated by an independent 4,000,000-call conservation fuzz that held, a divergent heretic pass (0 surviving frame-breaks), and the project's own 33 Certora proofs. A hardened, formally-verified isolated-lending primitive.

Per-finding details are withheld from this public summary.

Titles, code locations, PoCs and remediations are provided in the full report.

Full report - 101 pages - available on request

Findings - PoCs - Certora coverage map - attack-chain ledger - remediation specs

Contact

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Credit: Kairos Lab Security Research. DRAFT - confidential until disclosure.